Speaking Indian Languages, Missing Indian Culture: A Taxonomy of LLM Cultural Competence Gaps

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Abstract

Cultural understanding remains one of the most underexplored dimensions of large language model (LLM) evaluation and development, particularly for linguistically diverse regions such as the Indian subcontinent. This survey organizes the landscape of cultural understanding in LLMs around four complementary dimensions: cultural grounding, cultural reasoning, cultural-semantic intelligence, and cultural safety and representation. With a focused examination of Indic languages, we synthesize findings across datasets, benchmarks, model performance, and approaches to improving cultural competence. We cover major resources including SANSKRITI, MILU, DRISHTIKON, VIRAASAT, DOSA, DIWALI, and BhashaBench, and document a central finding: even regional Indic LLMs trained on local languages fail to align with Indian cultural values. We identify critical gaps in low-resource and endangered language coverage, document the limitations of translation-based approaches, and argue that linguistic fluency and genuine cultural alignment are fundamentally distinct properties that require separate, deliberate investment.

1 Introduction

Large language models have achieved remarkable performance across a wide range of natural language processing tasks, yet their cultural understanding remains shallow and unevenly distributed across the world's languages and communities (Seth et al., 2024). True cultural competence requires exposure to culture-specific concepts, values, practices, and artifacts that are often absent or underrepresented in the web-crawled corpora that dominate model pretraining (Nyandwi et al., 2025). For India, with over 1.4 billion people speaking more than 120 languages and around 19,500 dialects across 28 states (Verma et al., 2025), the need for culturally appropriate benchmarks is especially urgent. Models deployed in South Asian contexts regularly misinterpret culturally grounded queries, fail to recognize regionally specific artifacts, and default to Western-centric assumptions when handling ambiguous inputs (Nyandwi et al., 2025).

A particularly striking finding has emerged from recent work: regional Indic LLMs trained specifically for Indian languages do not align better with Indian cultural values than global models. In fact, a U.S. respondent is a closer proxy for Indian values than any Indic model tested, and regional finetuning can even degrade existing cultural knowledge (Agarwal et al., 2026). This reveals that multilingual fluency and cultural alignment are fundamentally distinct properties that do not follow automatically from each other. Figure 1 illustrates the scale of the underlying data imbalance: English dominates cultural LLM research while most Indic languages appear in fewer than 20 of the 59 surveyed works.

This survey makes three contributions beyond synthesis. First, we propose a four-dimensional taxonomy of cultural competence in LLMs: *Cultural Grounding* (what communities know, value, and practice), *Cultural Reasoning* (how communities interpret, infer, and decide), *Cultural-Semantic Intelligence* (how culture encodes meaning through stories, metaphors, and concepts), and *Cultural Safety and Representation* (fairness, harms, stereotypes, and inclusion). This taxonomy reveals that prior work is heavily concentrated in cultural grounding while cultural-semantic intelligence and systematic safety evaluation remain comparatively underexplored. Second, we focus specifically on Indic languages, enabling region-level analysis that global surveys cannot support. Third, we document and contextualize the fluency-alignment gap: the finding that cultural competence requires explicit design choices well beyond linguistic coverage.

2 Related Work

Several recent surveys address cultural understanding in LLMs. Adilazuarda et al. (2024) survey more than 90 papers on cultural representation in LLMs, organizing culture into semantic proxies and demographic proxies, and find that only values and objectives have been studied while semantic domains and cultural "aboutness" remain largely unexplored. Pawar et al. (2025) offer a broader survey covering cultural awareness across text and multimodal LLMs, examining cross-cultural dataset construction, cultural inclusion strategies, and benchmarking methods.

Our survey differs from these in three ways. We focus specifically on Indic languages and Indian cultural contexts rather than adopting a fully global scope, enabling analysis of sub-regional variation (e.g., regional vs. national commonsense) that global surveys cannot support. We organize around four dimensions chosen to highlight where Indic research is concentrated and where it is sparse, rather than adopting generic fairness categories or value-dimension frameworks. And we foreground the fluency-alignment distinction–the finding that regional Indic LLMs fail to align with Indian values despite linguistic coverage–which neither prior survey addresses directly. Our contributions are thus precision and Indic focus rather than broad coverage.

3 Definitions and Frameworks

Culture encompasses material elements such as food, dress, housing, and ornaments alongside non-material components including beliefs, norms, values, and moral systems (Sahoo et al., 2025). In NLP research for Indic contexts, Cultural Specific Items (CSIs) have emerged as central proxies for evaluating cultural competence, representing concepts, objects, and customs that carry significant cultural importance. Examples include Mekhela Chador (a traditional Assamese garment), Pongal (a harvest festival and dish from Tamil Nadu), and Darjeeling Tea (Sahoo et al., 2025). This material-centered operationalization has practical advantages for dataset construction but risks underrepresenting oral traditions, kinship structures, and cosmological beliefs that are harder to anchor to concrete artifacts.

A long-standing critique in behavioral science is that research relying on Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies overgeneralizes to non-WEIRD contexts. In LLM evaluation this manifests as English-language benchmarks, evaluation designs reflecting Western assumptions about cultural competence, and metrics that privilege written, explicit cultural knowledge over oral traditions and embodied practice. Seth et al. (2024) explicitly respond to this gap by arguing that for LLMs to have global acceptability, models must go beyond artifacts that have been absorbed into mainstream discourse and encode the knowledge present within communities that are not well represented on the Web.

Operationalizing culture for NLP evaluation requires choosing among distinct approaches. Material-centered approaches work through CSIs and are easy to benchmark. Value-dimension approaches, such as Hofstede's cultural dimensions (Hofstede and Minkov, 2013) used in Feng et al. (2025), provide systematic cross-cultural comparison but impose external taxonomies. Community-centered participatory approaches, exemplified by DOSA (Seth et al., 2024), capture lived cultural meaning through bottom-up, community-engaged data collection where participants use the film *Don* as a cultural anchor for communicating about Benaras rather than relying on official designations. Knowledge-graph approaches, used in VIRAASAT (Surana et al., 2026), enable structured multi-hop reasoning about cultural facts. Each approach captures different facets of culture; robust evaluation requires triangulating across multiple methods rather than treating any single proxy as sufficient.

Who decides what counts as culturally correct is not a neutral question. Traditional benchmarks rely on expert annotation. Participatory approaches like DOSA engage community members directly. When a benchmark asks whether a response to a cultural query is appropriate, these two approaches often disagree, and the choice of authority embeds assumptions about whose cultural knowledge is considered authoritative.

4 Cultural Grounding

Cultural grounding concerns what communities know, value, and practice. Research in this dimension asks whether LLMs can correctly recognize, identify, and reason about culturally specific facts, artifacts, and traditions.

4.1 Western-Centric Bias in LLMs

A consistent finding across studies is that LLMs exhibit strong Western-centric cultural bias. Tao et al. (2024) examined cultural bias across five GPT models by calculating cultural distance from the Inglehart-Welzel cultural map, finding that GPT-4o's cultural values are closest to Finland (distance 0.20), Andorra (0.21), and Netherlands (0.45), while most distant from Jordan (4.10), Libya (4.00), and Ghana (3.95). Cao et al. (2023) similarly found that LLMs overrepresent English-speaking and Protestant European cultural perspectives, and that English prompts reduce variance in model responses while biasing them toward American culture. Parameter scaling and better preference tuning improve performance, but gains are more pronounced for English and European cultural norms than for African-Islamic cultures (Rao et al., 2025).

The performance gap across cultures is stark. BLEnD (Myung et al., 2025) demonstrates that the average performance of all tested models on questions about United States culture in English is 79.22%, but drops to only 12.18% when asked about Ethiopian culture in Amharic, across 52,557 question-answer pairs spanning 16 countries in 13 languages. Global MMLU (Singh et al., 2025) shows that state-of-the-art models excel primarily by learning Western-centric concepts, and that model rankings change when evaluated on questions explicitly marked as culturally sensitive. Evaluations on CVQA (Romero et al., 2024), GIMMICK (Schneider et al., 2025), and CulturalBench (Chiu et al., 2025) confirm the same pattern: strong Western bias and weak performance on non-Western cultural content across both text and multimodal settings.

4.2 Indic Language Benchmarks

SANSKRITI (Maji et al., 2025a) provides 21,853 questions spanning 28 Indian states and 8 union territories, covering 16 cultural attributes including rituals, history, cuisine, dance, music, festivals, religion, and medicine. DRISHTIKON (Maji et al., 2025b) extends evaluation to the multimodal setting with over 64,000 aligned text-image pairs across 15 Indian languages and all states and union territories. DIWALI (Sahoo et al., 2025) captures approximately 8,817 culture-specific concepts across 17 facets and 36 sub-regional contexts, verified against Government of India portals and state tourism websites. MILU (Verma et al., 2025) covers 11 Indic languages with around 79,000 questions drawn from competitive examinations. For domain-specific grounding, BhashaBench V1 (Devane et al., 2025) provides 74,166 curated question-answer pairs covering Agriculture, Legal systems, Finance, and Ayurveda in English and Hindi. At the community level, DOSA (Seth et al., 2024) collects 615 social artifacts from 19 Indian geographic subcultures through participatory research with 260 participants, ensuring that artifacts and their descriptions align with the shared sensibilities of community members rather than external taxonomies.

4.3 Model Performance and Key Findings

GPT-4o achieves the highest average accuracy at 74% on MILU, with open-source models showing substantially greater variation (Verma et al., 2025). Open multilingual models outperform language-specific fine-tuned models, which perform only marginally better than random baselines. Domain analysis reveals that models perform significantly worse on culturally grounded areas such as Arts and Humanities and Law and Governance than on general STEM subjects. BhashaBench V1 shows that models are strongest on Legal content and weakest on Ayurveda, with English consistently outperforming Hindi across all domains (Devane et al., 2025). Benchmarking Hindi LLMs (Kamath et al., 2025) confirms that direct translation fails to preserve linguistic subtleties: MT-Bench-Hi scores range from 3.44 to 8.70, GSM8K-Hi from 26.88% to 93.41%, and IFEval-Hi from 36.08% to 73.86% across tested models.

A deeper pattern emerges when language of prompting is considered. Cultural commonsense knowledge becomes inaccessible when queried in languages other than English, and asking about a country's culture in English frequently outperforms asking in the local language (Shen et al., 2024). The most troubling finding comes from Agarwal et al. (2026), who evaluate six Indic and six global LLMs on nationally representative surveys and community-sourced QA datasets. Indic models do not align better with Indian norms than global models, and regional fine-tuning can degrade alignment rather than improve it. The authors attribute this to the fundamental scarcity of culturally grounded pretraining data and call for native, community-authored corpora as a prerequisite for genuinely aligned Indic LLMs.

Chhikara et al. (2025) find that the highest accu-

racy across all models on Indian subculture tasks reaches only 41.6% in vanilla prompting. When automated context enrichment is applied, Llama achieves 91.6%, GPT-4o reaches 83.3%, Mixtral and Gemini both record 75%, and GPT-4o-mini achieves 41.6%, with performance deteriorating in most cases when prompted in a regional language rather than English. Cultural alignment fails even at the retrieval level: across 12 Indian languages, the best embedding model achieves only 27.4% Recall@1 on persona-to-instruction matching, indicating that retrieving culturally appropriate responses for a given user persona remains an unsolved problem (Shah et al., 2026). At the sub-language level, Sharma et al. (2026) show that GPT-4o and Gemini 2.5 perform poorly on dialect classification across 11 dialects of Hindi and Odia (19.6% F1 baseline), while fine-tuned transformer models improve this to 89.8%, highlighting that dialectal variation within a single language presents challenges as significant as cross-language coverage.

Having established the scale and nature of cultural grounding failures, we now turn to cultural reasoning, where models must go beyond recognition to perform inference and adaptation.

5 Cultural Reasoning

Cultural reasoning concerns how communities interpret, infer, and decide. Research in this dimension asks whether LLMs can perform multi-hop inference, contextual adaptation, and norm-aware reasoning that reflects the implicit logic of specific cultural communities.

5.1 Regional Cultural Commonsense

A foundational assumption in most cultural benchmarks is that nations are culturally homogeneous, but cultural commonsense in India is predominantly regional, not national. INDICA (Madhusudan et al., 2026) collects human-annotated answers from five Indian regions across 515 questions spanning 8 domains of everyday life, yielding 1,630 region-specific question-answer pairs. Only 39.4% of questions elicit agreement across all five regions. Domain-level variation is substantial: Traffic and Transport Behavior shows the highest agreement (22.6%), reflecting nationwide standardization, while Festivals and Rituals (1.8%) and Food Processing and Consumption (6.0%) show the lowest, reflecting strong regional traditions. Even Education achieves only 13.8% despite national curricula, showing that regional practices persist even in nominally standardized domains. These differences are substantive rather than linguistic: harvest festival games yield Jallikattu in South India versus kite flying in Central India. Models achieve only 13.4% to 20.9% accuracy on region-specific questions and exhibit geographic bias, over-selecting Central and North India as cultural defaults.

5.2 Multi-Hop and Cross-Cultural Reasoning

VIRAASAT (Surana et al., 2026) evaluates multi-hop cultural reasoning with over 700 expert-curated cultural artifacts spanning 13 attributes across all 28 Indian states and 8 union territories, generating more than 3,200 complex questions requiring chained cultural reasoning. The knowledge graph underlying VIRAASAT represents hierarchical relationships between state, sub-region, community, and behavioral norm, enabling questions that require multi-step inference rather than simple fact lookup. Fine-tuning on Chain-of-Thought traces alone fails to ground and synthesize low-probability cultural facts. The proposed Symbolic Chain-of-Manipulation framework, which teaches the model to simulate atomic Knowledge Graph manipulations internally, outperforms standard CoT baselines by up to 20%.

Cross-cultural reasoning is evaluated by XCR-Bench (Kabir et al., 2026), which contains 4.9k parallel sentences and 1,098 unique CSIs across Chinese, Arabic, and Bengali settings, grounded in Newmark's CSI framework and Hall's Triad of Culture. Performance declines significantly on culturally sensitive categories and deeper cultural levels ($p < 0.005$, 8/8 models), and adaptation quality varies systematically across target cultures and Bengali regional variants, revealing encoded regional and ethno-religious biases. CultureForest (Ye et al., 2026) frames cultural norm grounded reasoning as the core evaluation task, requiring models to determine appropriate behavior from culturally grounded premises in a Verifiable and Attributable manner. Global PIQA (Chang et al., 2026) extends commonsense evaluation to 100+ languages and cultures.

5.3 Cultural Adaptation Evaluation

The Adaptation Score introduced by Sahoo et al. (2025) measures cultural concept replacement quality through Exact Match and Fuzzy Match metrics, evaluated by diverse socio-demographic raters from different Indian states. A key finding is that

models perform surface-level concept replacement but fail at deeper adaptation. When adapting a scenario involving Tuesday to one involving Diwali, a model correctly swaps the concept but continues to describe selling CDs rather than transitioning to selling festival decorations, demonstrating failure to establish meaningful cultural aboutness (Sahoo et al., 2025).

5.4 Cross-Lingual Transfer

CulturalPangea demonstrates that culturally grounded training scales to underrepresented languages without sacrificing performance on higher-resource ones (Nyandwi et al., 2025). Absolute accuracy gains on ALMBench include 15.0 percentage points on Sinhala, 10.9 on Hebrew, and 9.1 on Irish, with consistent improvements across Tamil (+6.3), Amharic (+5.3), Bengali (+4.8), and Telugu (+4.2), as shown in Figure 2. CulFiT (Feng et al., 2025) demonstrates complementary gains through fine-grained culturally aware reward modeling. Using Hofstede's cultural dimensions as evaluation, CulFiT reduces cultural distance for Llama3.1 from 174.83 to 135.24 and for Qwen2.5 from 157.41 to 140.32, outperforming GPT-4o on cultural value alignment despite GPT-4o's superiority on fundamental tasks such as math and coding. Improvements are most pronounced for low-resource regions, with Sundanese in West Java increasing from 12.43 to 20.13 and Amharic in Ethiopia from 8.26 to 12.34.

6 Cultural-Semantic Intelligence

Cultural-semantic intelligence concerns how culture encodes meaning through stories, metaphors, proverbs, folklore, and conceptual systems. Meaning in this dimension is often implicit, context-dependent, and tied to shared community history in ways that resist simple translation.

6.1 Figurative Language and Proverbs

Figurative language carries implicit cultural references that vary considerably across communities (Kabra et al., 2023). A Nepali proverb referencing Shankar encodes beliefs in divine protection through Hindu deity recognition; for models unfamiliar with that cultural context, the meaning is entirely lost (Thapa et al., 2025). Analysis across Hindi, Indonesian, Javanese, Kannada, Sundanese, Swahili, and Yoruba shows that each language relies on cultural and regional concepts for figurative expression, with the highest overlap occurring between languages from the same region (Kabra et al., 2023). Machine translation of figurative datasets exposes this fragility directly: as shown in Figure 3, 40% of Hindi examples are incorrect after translation and 20% are culturally irrelevant, with Japanese showing 21% incorrect and 17% irrelevant. Translation errors in this domain are not merely linguistic but epistemic, replacing culturally specific meanings with globally dominant defaults (Kabra et al., 2023).

Arabic proverbs illustrate a related challenge at the sub-language level. Magdy et al. (2025) evaluate LLMs on multidialectal Arabic proverbs spanning Palestinian, Syrian, Kuwaiti, Omani, and other dialects, finding that while LLMs can generate idiomatically accurate translations, they struggle with culturally nuanced and contextually relevant explanations.

6.2 Multimodal Cultural Meaning

Cultural meaning extends beyond text. Yang et al. (2025) document cross-cultural differences in multimodal metaphor: the phrase "someone is a dinosaur" means old-fashioned in English but conveys ugly in Chinese, while Western cultures describe intoxication through animal metaphors such as "drunk as a skunk" whereas Chinese culture uses mud as in "drunk as mud." CultureBank (Shi et al., 2024) provides 23,000 cultural descriptors sourced from TikTok and Reddit, built on users' self-narratives with diverse community views to allow flexible interpretation of cultural knowledge and avoid encoding a single dominant perspective.

6.3 Cultural Concept Systems

DIWALI (Sahoo et al., 2025) captures cultural-semantic knowledge across 17 facets including naming conventions, languages and dialects, and tribal traditions alongside tangible categories such as food and clothing. The failure mode here is mundane but persistent: asking a model to recommend a herb to plant in the backyard produces Western herbs like thyme and rosemary rather than culturally familiar options such as tulsi, pudina, and curry leaves (Rachamalla et al., 2025). Models replace indigenous concepts with globally dominant equivalents not only in cultural knowledge tasks but in everyday practical reasoning.

While cultural semantics concerns how meaning is encoded, cultural safety concerns the impact of meaning failures when models produce outputs that

harm or misrepresent communities.

7 Cultural Safety and Representation

Cultural safety and representation concerns fairness, harms, stereotypes, and inclusion. Research in this dimension asks whether LLMs produce outputs that are appropriate, non-discriminatory, and sensitive to cultural context.

7.1 Stereotyping and Bias in Indian Contexts

LLMs consistently produce stereotypical outputs in Indian contexts. IndiBias (Sahoo et al., 2024) documents associations pairing Brahmin identity with Elitist, Kshatriya with Power-hungry, Muslim with Terrorist, and Shudra with Subservience, illustrating how caste and religious biases from skewed training data propagate into model outputs. Synthetic data generation for Indic pretraining amplifies rather than mitigates these patterns: BhashaKritika (Manoj et al., 2025) reports caste bias effect sizes ranging from 0.56 to 1.09, gender bias peaking at 1.58 in story-style generation, race bias peaking at 1.51 in blogpost style, and religion bias ranging from 1.39 to 1.73 across styles, reflecting strong Hindu-Muslim stereotypes embedded in synthetic corpora.

IndRegBias (Panda et al., 2026) focuses specifically on regional bias, collecting 25,000 comments from Reddit and YouTube on regional issues in India. Zero-shot and few-shot approaches show lower accuracy in detecting regional biases across most LLMs, while fine-tuning significantly enhances performance in detecting both the presence and severity of regional bias.

7.2 Multilingual Safety

Safety behavior varies substantially across languages. IndicSafe (Pattnayak and Chowdhuri, 2026) presents the first systematic evaluation of LLM safety across 12 Indic languages, using 6,000 culturally grounded prompts spanning caste, religion, gender, health, and politics assessed across 10 leading LLMs. Cross-language agreement is just 12.8%, implying that identical prompts yield different safety judgments in nearly 90% of cases when the language changes. Across all 60,000 model generations, SAFE responses constituted 50.5%, UNSAFE 18.7%, REFUSAL 18.4%, and AMBIGUOUS 12.4%.

Addressing the lack of automated moderation tools for low-resource Indic languages, UnityAI-Guard (Beniwal et al., 2025) introduces a binary toxicity classification framework targeting seven Indic languages written in Brahmic scripts. Trained on 567k instances with 30k manually verified test examples, the system achieves an average F1-score of 84.23% and provides public API access to support broader multilingual content moderation.

XL-SafetyBench (Choi et al., 2026) evaluates safety and cultural sensitivity across 10 countries and 5,500 test cases, covering country-specific jailbreak scenarios and hidden cultural mistakes across six categories: Symbolic Taboos and Gift-Giving, Food and Dietary Law, Death and Funeral Practices, Daily Life and Public Conduct, Hierarchy and Social Respect, and Legal Landmines. A key finding is that strong safety performance does not imply strong cultural understanding, and local language models are not automatically culturally aware despite being trained on local text.

7.3 Cultural Representation in Generated Content

TALES (Bhagat et al., 2026) examines cultural representations in LLM-generated stories, finding that models draw heavily on cultural markers visible to outsiders. Generated stories feel like portrayals of what a culture is known for rather than authentic lived representations, with systematic patterns of oversimplification that flatten cultural nuance.

7.4 Evaluation Framework Gaps

Current evaluation frameworks face persistent structural limitations beyond dataset coverage (Sinha et al., 2026). Many multilingual benchmarks are simply translations of English-language benchmarks, leading to loss of linguistic nuance, cultural meaning, and contextual relevance. These limitations are especially consequential in high-stakes domains such as healthcare, where Hamna et al. (2026) show that community-centered evaluation methods are necessary to capture local healthcare practices and linguistic realities across Hindi, Kannada, and Malayalam. Most frameworks also favor dominant dialects even within a single language family, making genuinely non-English-centric evaluation difficult in practice.

These documented gaps in grounding, reasoning, semantics, and safety have motivated a range of approaches to improvement. The section below examines what shows the most promise.

8 Approaches to Improving Cultural Competence

8.1 Culturally Grounded Training Data

The IndicLLMSuite (Khan et al., 2024) provides a blueprint for culturally anchored pretraining, encompassing the SANGRAHA dataset of 251 billion tokens across 22 languages drawn from curated URLs, existing multilingual corpora, and large-scale translations with custom tokenizers supporting English and all 15 major Indic languages. The Pragyaan framework (Rachamalla et al., 2025) extends this to post-training, combining translation with targeted human refinement across ten Indian languages covering 13 broad and 56 sub-categories. Annotators engage in three levels of work: correcting linguistic errors, adapting expressions to reflect Indian cultural norms, and producing culturally substituted variants where direct translation is inadequate. The distinction between literal translation and cultural substitution is critical: models trained only on translated data never encounter scenarios where cultural concepts are incommensurable and require genuine cultural reasoning rather than code-switching. Modular pipelines for generating synthetic Indic multilingual corpora (Manoj et al., 2025) complement these efforts by ensuring synthetic outputs are factually grounded and topically diverse.

CultureLLM (Li et al., 2024) takes a different approach, using World Values Survey data as cultural seed knowledge expanded through semantic data augmentation before fine-tuning across nine cultures, improving performance by 8.1% over GPT-3.5 and 9.5% over Gemini on culture-related tasks.

8.2 Knowledge-Graph-Based and Multimodal Approaches

CulturalPangea (Nyandwi et al., 2025) leverages Wikidata's multilingual knowledge graph through a five-stage pipeline, producing 22 million VQA pairs spanning 42 countries and 39 languages, enabling a model that outperforms prior open models by an average of 5.0% on culture-focused multimodal benchmarks. The knowledge graph approach is particularly well-suited to Indian cultural contexts because it can represent hierarchical relationships (state, sub-region, community, behavioral norm) and enable multi-hop reasoning about cultural implications rather than surface-level fact lookup. VIRAASAT (Surana et al., 2026) demonstrates this advantage specifically for Indian cultural reasoning. Pangea (Yue et al., 2025) extends the multimodal approach to a fully open multilingual LLM supporting 39 languages. Regional multimodal adaptation is examined by Cahyawijaya et al. (2026) for Southeast Asian languages, finding that models require culturally grounded visual and textual data to adapt to regional cultural contexts.

8.3 Indic-Specific LLMs

Several efforts focus on building LLMs specifically for Indian languages. Krutrim (Kumar et al., 2025) introduces a novel tokenization strategy trained on petabyte-scale data across 12 Indian languages. Airavata (Gala et al., 2024) introduces a Hindi instruction-tuned LLM combining translated instruction data with cultural adaptation. A critical caution from Agarwal et al. (2026) is that linguistic coverage alone does not produce cultural alignment. Regional pretraining on web text from India absorbs regional stereotypes and biased narratives endemic in online discourse, and without explicit cultural grounding in data curation and curriculum design, these models can entrench rather than mitigate cultural problems. The path to culturally competent Indic LLMs is not simply linguistic.

8.4 Tokenizer and Infrastructure Optimization

MUTANT (Rana et al., 2026) demonstrates that vocabulary size, data distribution, language-specific pre-tokenization, and multi-stage learning can be systematically optimized for Indic languages. Tokenizer design that accounts for morphological richness and script diversity is essential to ensure that model architectures do not penalize Indic languages at the infrastructure level before any cultural content is even processed.

8.5 Endangered Language Coverage

ELR-1000 (Joshi et al., 2025) presents 1,060 traditional recipes crowdsourced from rural communities spanning 10 endangered languages, revealing both the feasibility and the depth of the challenge: even the best models struggle with low-resource, culturally specific language. Languages with no existing LLMs cannot leverage model-generated data for instruction fine-tuning, making initial resource creation especially difficult (Khan et al., 2024). Current large-scale benchmarks remain largely confined to high-resource Indic languages, leaving endangered and tribal languages effectively unrepresented.

8.6 Integration and Limitations of Existing Approaches

While the approaches above show individual promise, their integration reveals persistent limitations. Culturally grounded training data is most effective when combined with community-centered evaluation, yet few studies employ both. Knowledge graphs work well for structured cultural facts but struggle with subjective cultural meaning. Indic-specific LLMs offer linguistic advantages but require cultural curation to avoid amplifying biases. Most critically, all existing approaches operate downstream of the cultural gap: they improve models after the fact rather than investing upstream in participatory data collection, community-led benchmark design, and evaluation metrics that communities themselves help define.

9 Key Datasets

Table 1 and Figure 4 summarize the major datasets for Indic cultural understanding evaluation. The variation in scale reflects complementary design philosophies: large-scale MCQ benchmarks like MILU (79,000 questions) and DRISHTIKON (64,000+ pairs) enable broad coverage, while community-generated resources like DOSA (615 artifacts) and knowledge-graph-driven VIRAASAT (3,200+ questions) prioritize qualitative depth.

10 Conclusion

Across all four dimensions, a consistent finding emerges: linguistic fluency in an Indic language does not imply cultural alignment with the communities that speak it. Even regional LLMs trained specifically for Indian languages reflect American rather than Indian values, and regional fine-tuning often fails to close this gap (Agarwal et al., 2026). Cultural grounding benchmarks reveal persistent performance gaps in culturally relevant domains. Cultural reasoning evaluations show that surface-level concept recognition does not translate to deep cultural adaptation. Cultural-semantic intelligence is systematically degraded by machine translation. Cultural safety evaluation reveals that safety judgments are not language-invariant, with identical prompts yielding different outcomes in nearly 90% of cases when the language changes (Pattnayak and Chowdhuri, 2026).

The evidence points toward a common root cause: the absence of natively sourced, culturally grounded training data authored by and for the communities in question. Scaling up translated data or region-specific fine-tuning is insufficient (Nyandwi et al., 2025). What is needed is upstream investment in community-centered data collection across all 22 constitutionally recognized Indian languages, spanning registers from classical to colloquial and covering cultural phenomena at the level of sub-regional communities and their little traditions rather than only the nationally visible markers of culture.

Three research priorities follow from this survey. First, scaling participatory evaluation beyond the current handful of projects to cover sub-regional and dialectal variation, which INDICA shows is as significant a challenge as cross-language coverage. Second, bridging the fluency-alignment gap through architectures and curricula that treat cultural grounding as a first-class design objective rather than a post-hoc addition. Third, shifting evaluation authority toward communities, so that cultural benchmarks measure what communities themselves consider culturally competent behavior rather than what external annotators and official sources define as correct. These are expensive and complex commitments, but the evidence accumulated across the 59 works surveyed here makes clear that incremental improvements to existing approaches will not be sufficient.

Limitations and Research Ethics

This survey covers works published or made available through mid-2026 and may not capture the most recent developments in this rapidly evolving area. Our analysis emphasizes Hindi, Tamil, Telugu, Kannada, Bengali, and Malayalam, which account for the majority of Indic language papers surveyed, and is necessarily sparse for endangered and tribal languages.

The four-pillar taxonomy represents one useful decomposition of cultural understanding but not the only one. By organizing around benchmarks and datasets, this survey may overemphasize phenomena that are easy to measure while underemphasizing harder-to-measure dimensions such as authentic cultural meaning-making and community satisfaction with model outputs. Several claims about dataset statistics rely on author-reported figures that we have not independently verified.

Benchmarks for cultural understanding embed hidden epistemological choices about what counts as cultural knowledge, which cultural phenomena

Dataset	Coverage	Scale	Key Focus	Year
DOSA (Seth et al., 2024)	19 Indian subcultures	615 artifacts	Social artifacts, participatory research	2024
SANSKRITI (Maji et al., 2025a)	28 states + 8 UTs, 16 attributes	21,853 questions	Cultural knowledge across India	2025
DIWALI (Sahoo et al., 2025)	36 sub-regions, 17 facets	∼8,817 concepts	Cultural text adaptation	2025
VIRAASAT (Surana et al., 2026)	28 states + 8 UTs, 13 attributes	3,200+ questions	Multi-hop cultural reasoning	2026
DRISHTIKON (Maji et al., 2025b)	15 languages, all states/UTs	64,000+ pairs	Multimodal cultural understanding	2025
MILU (Verma et al., 2025)	11 Indic languages	79,000 questions	Multilingual cultural knowledge	2025
IndicParam (Maheshwari et al., 2025)	11 Indic languages	13,207 questions	Low-resource language evaluation	2025
BhashaBench V1 (Devane et al., 2025)	English & Hindi, 4 domains	74,166 QA pairs	Domain-specific Indian knowledge	2025
INDICA (Madhusudan et al., 2026)	5 Indian regions, 8 domains	1,630 pairs	Regional cultural commonsense	2026
IndicSafe (Pattnayak and Chowdhuri, 2026)	12 Indic languages, 10 LLMs	6,000 prompts	Multilingual safety evaluation	2026

Table 1: Major cultural understanding datasets for Indic contexts. Scale is reported as number of instances depending on dataset type.

are visible and measurable, and who holds authority to define culturally correct behavior. Models trained to minimize error on cultural benchmarks may learn to amplify cultural stereotypes if benchmarks themselves encode stereotypical assumptions. This survey does not resolve these tensions but emphasizes that practitioners should treat high benchmark scores as evidence of strong performance on expert-designed, declarative cultural knowledge tasks rather than as evidence of genuine cultural understanding.

Most datasets mentioned in this survey are publicly available, though access conditions vary across resources. This survey is written in English by researchers who may not be members of all communities whose cultures are surveyed, reflecting broader power asymmetries in NLP research. We encourage readers to seek out work by authors from Indic-speaking communities and to support participatory research models where community members are co-investigators and decision-makers.

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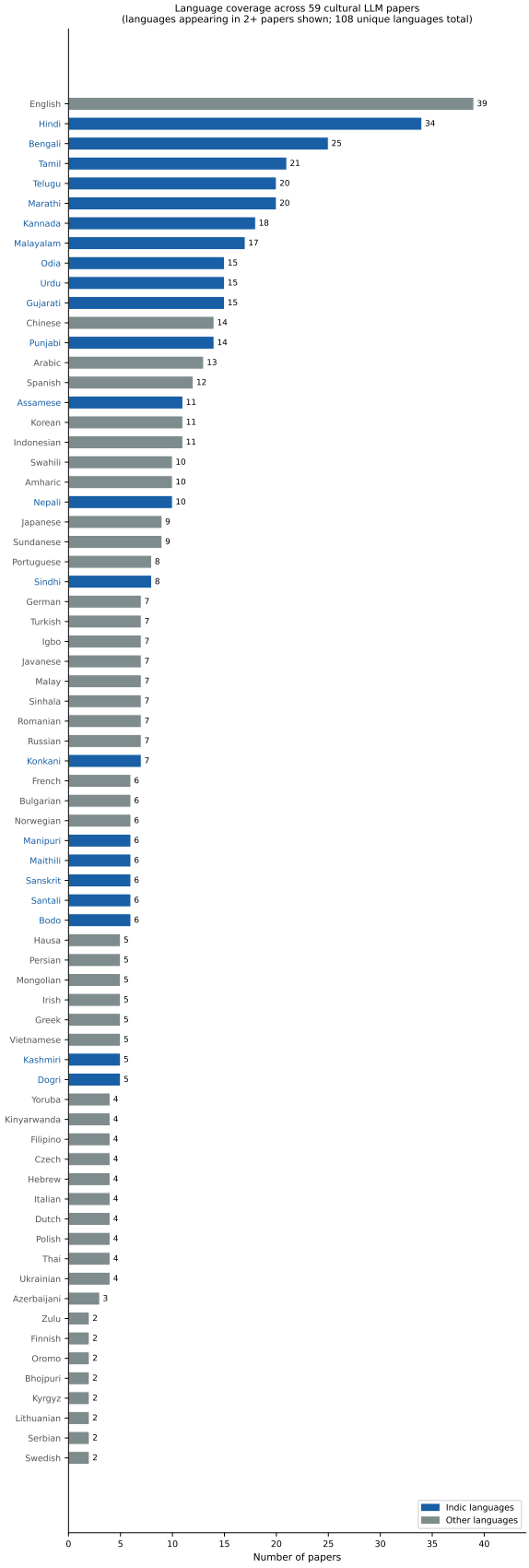


Figure 1: Number of surveyed cultural LLM papers covering each language across 59 works. Indic languages are highlighted in blue. Despite India having 22 constitutionally recognized languages, most appear in fewer than 20 papers. English dominates at 39 papers, with 108 unique languages appearing across all surveyed works.

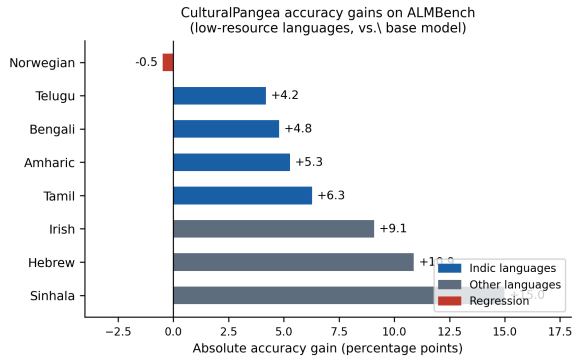


Figure 2: Absolute accuracy gains of CulturalPangea on ALMBench compared to the base model across low-resource languages (Nyandwi et al., 2025). Indic languages (blue) show consistent gains of 4.2 to 6.3 percentage points, confirming that culturally grounded training scales effectively to underrepresented languages.

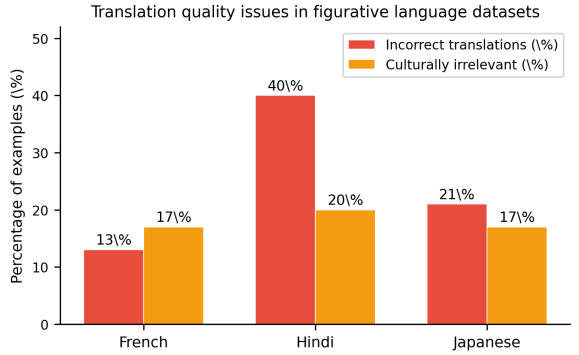


Figure 3: Percentage of incorrect and culturally irrelevant examples in machine-translated figurative language datasets across French, Hindi, and Japanese (Kabra et al., 2023). Hindi shows the highest error rate at 40%, more than three times that of French, illustrating how translation quality degrades as cultural distance from English increases.

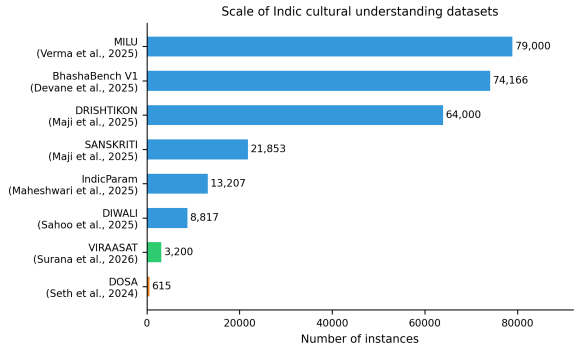


Figure 4: Scale of major Indic cultural understanding datasets by number of instances. MILU and DRISHTIKON are the largest at roughly 79,000 and 64,000 instances respectively, while community-generated datasets like DOSA (615) and knowledge-graph-based VIRAASAT (3,200+) are smaller but qualitatively richer.